Lab day 10 November: results

34870 Electroacoustics, loudspeaker project, Group 10 (System D)

Data: `lab-day/data/` (clean copies) and `raw data/` (lab-PC files). Figures: `analysis/results_labday.m`. Log: `LOG_2026-11-10.md`.

In one paragraph

Once the setup was right, the built system does what the design said it would. The woofer and the mid/tweeter channel are balanced to within 0.1 dB, the two crossovers are *more* in phase than the model predicted, and the measurements repeat to within 0.16 dB half an hour later. The one real deviation is tonal: the **tweeter is about 2.7 dB too quiet** and the midrange about 1.5 dB too loud against the model. That widens the on-axis spread from the predicted 4.0 dB to 7.1 dB (150 Hz–15 kHz). Changing one resistor fixes most of it: **Rp 3.3 Ω → 8.2 Ω** brings the spread down to about 5.3 dB. This is predicted from our own measurement, not only from the model.

Quantity	Result	Verdict
On-axis spread 150 Hz–15 kHz (1/6 oct)	7.1 dB (model 4.0 dB)	**to fix**
Woofer vs mid+tweeter level	−0.1 dB (after Out 1 = −5.8 dB)	**good**
Woofer / midrange / tweeter vs model	−0.1 / +1.5 / −2.7 dB	**to fix** (tweeter)
Woofer-flip notch	315 Hz, −32 dB (model 329 Hz, −21 dB)	**good**
Tweeter-flip notch	4161 Hz, −28.6 dB (model 3928 Hz, −14 dB)	**good**
Repeat after 31 min (R5 − R1)	median +0.07 dB, worst 0.16 dB	**good**
Sum check |woofer + mid/tw| vs |all|	worst 0.10 dB	**good**
Near field at 200 mm vs Lab D	within 1 dB over 91 % (cone), 86 % (vent)	**good**
Board impedance, 30°, Lab E placement re-check	not measured / not saved	**to fix**

1 What was measured

Settings: MiniDSP filters pasted from `data/dsp/MiniDSP_*.txt` (LR4, 325 Hz), **Out 1 = −5.8 dB** from R1 try 3 on (in tries 1 and 2 it was still at 0 dB: the design's −4.5 dB had not been typed in), Out 2 = 0 dB, amplifier about 45 %, placement as in Lab E (mic 2.44 m, 1.15 m high), vents plugged. The states were selected on the chamber's switchboard (woofer, midrange, tweeter each ON/OFF, as in Lab E). Since −5.8 dB instead of the design's −4.5 dB balanced the two channels, this lab's amplifier chain makes the woofer path only about 1.3 dB hotter than the model assumes.

Time	Run	Settings in the run	Used?
14:28	R1 try 1	woofer / mid+tw / all	no: woofer silent in "all"
15:02	R1 try 2	woofer / mid+tw / all	no: woofer 6.4 dB too loud (Out 1 still 0 dB)
15:17	**R1**	woofer / mid+tw / all	**yes**
15:24	R2 try 1	mid / tw / all	no: "all" not switched in time
15:31	**R2**	mid / tw / all	**yes**
15:40	**R3**	all / woofer inverted / tweeter swapped	**yes**
15:48	**R5**	repeat of R1	**yes**
16:28	**NF**	near field, cone and vent, one vent at 200 mm	**yes**

Not measured: R4 (30°), the impedances Z1–Z4. The Lab E re-check of the bare system was run but not saved to the stick.

2 The results

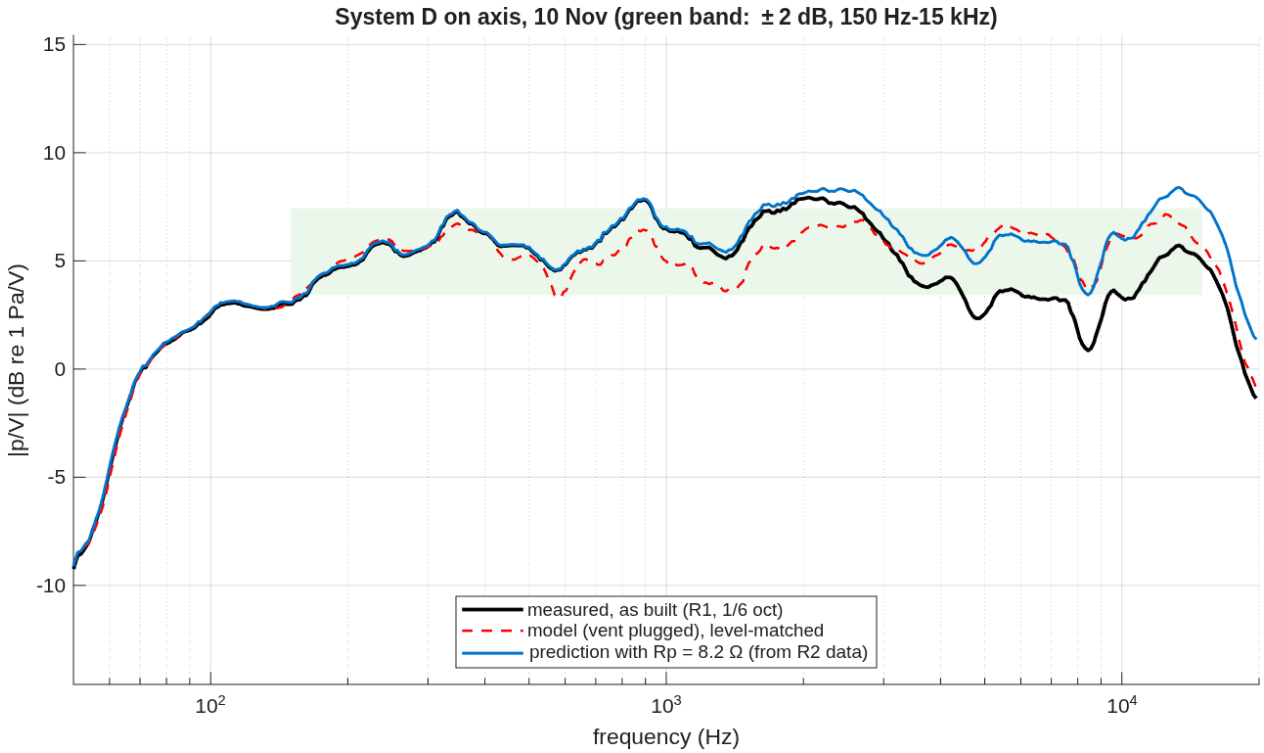


Figure 1: The whole system on axis. Black: as built. Red dashed: the model (Lab E data through our filters, vent plugged), shifted to the same level. Blue: what the same measurement becomes with Rp = 8.2 Ω (computed from run R2, see Section 4).

Figure 1 is the headline. Up to 3 kHz the measurement follows the model closely, with the same ripples at 350 Hz, 900 Hz and 2 kHz. Those ripples come from the drivers and the baffle, because they are already in the Lab E data. Between 700 Hz and 3 kHz we sit about 1.5 dB above the model; above 4 kHz about 3 dB below it. The 8.3 kHz dip is in the model too, so it belongs to the tweeter or the box, not to our crossover.

Figure 2 shows where that comes from. The woofer lands exactly on its model (−0.1 dB), and the two LR4 slopes cross at the right place and level. In the bottom panel the midrange runs 1.5 dB above its model and the tweeter 2.7 dB below it, so the mid–tweeter balance is off by about 4 dB. Three possible causes: the basket resistors and coils are not exactly their nominal values; the coils' DC resistance (not in the model) lowers the tweeter branch a little; and the tweeter may simply be less sensitive today than in Lab E. Thursday's placement step P3 (the tweeter through the board with the MiniDSP bypassed) and P3d (tweeter wired straight to the amplifier) separate these.

Figures 3 and 4 test the crossover *integration*. If the drivers are in phase at the crossover, flipping one of them makes the two cancel, which gives a deep notch. We got −32 dB at 315 Hz and −28.6 dB at 4.16 kHz, deeper than the model's −21 and −14 dB. The phase plot says the same thing directly. Woofer and mid/tweeter stay within about ±10° from 230 to 480 Hz. Midrange and tweeter go from +50° at 2.5 kHz through 0° at about 4.1 kHz, which is where the flip notch sits. So in the overlap region the units add up rather than fight, and there is no hole or peak at either crossover.

3 Why we can trust these measurements

- **They repeat.** R5 was the same measurement as R1, 31 minutes later, after three other runs and a lot of switching on the switchboard and polarity swapping. It lands within 0.16 dB everywhere

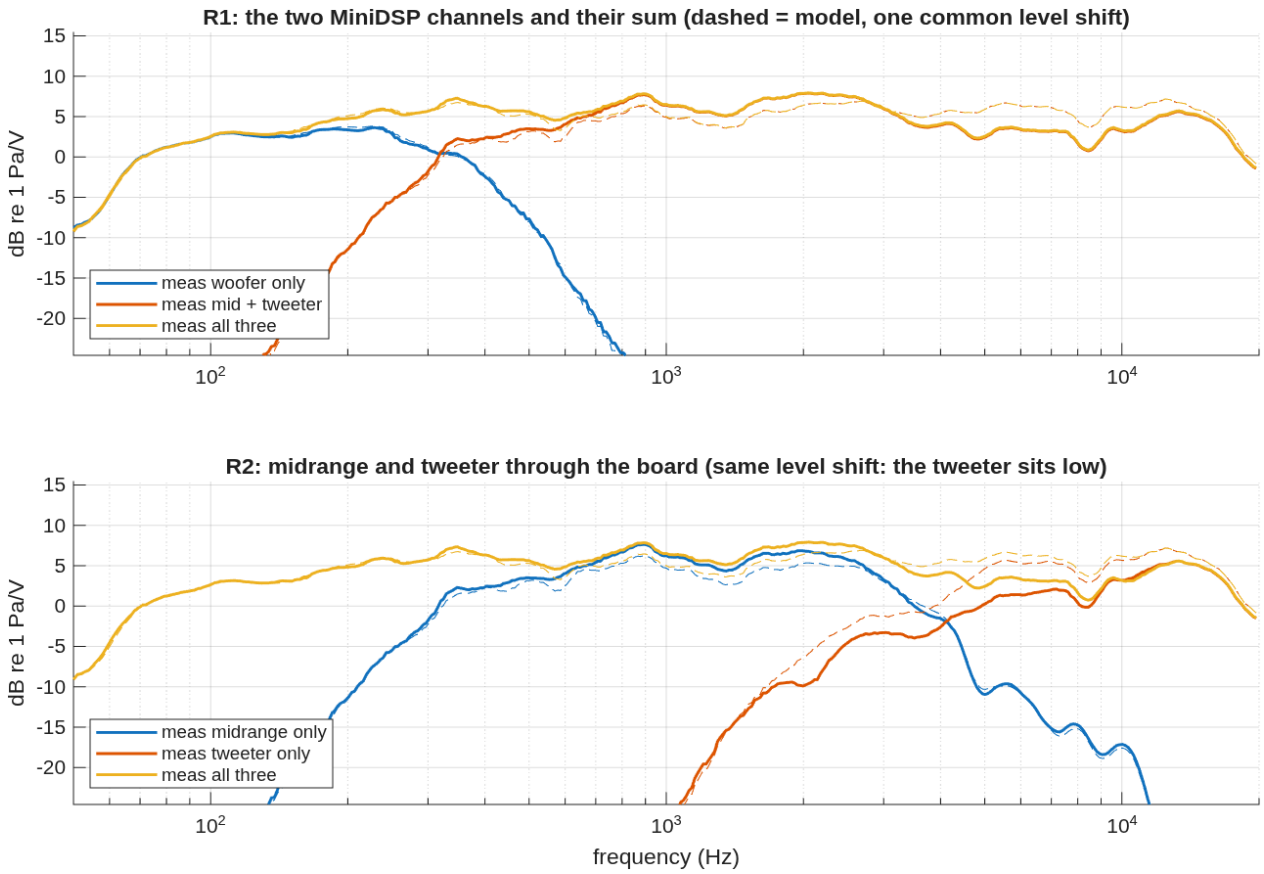


Figure 2: Top: the two MiniDSP channels and their sum. Bottom: midrange and tweeter separately through the board. All dashed model curves are shifted by *one* common amount, so differences in level between the units are real.

(Figure 5). Nothing moved and nothing drifted: not the microphone, the temperature or the amplifier.

- **They are self-consistent.** Within one run the woofer state plus the mid/tweeter state must add up, as complex numbers, to the "all" state. They do to within 0.10 dB. That can only happen if the system is linear, nothing was touched between the states, and the phase reference held.
- **They agree with independent earlier data.** The whole shape up to 3 kHz, including the small ripples, matches the model built from Lab E. The near field matches Lab D to within 1 dB over about 90 % of 10–200 Hz (Figure 6). The remaining differences are where the signal is tiny: the cone's own dip at the vent tuning (21 Hz), where the vent does the work, and the vent above 190 Hz. That also tells us the box (filling, plugs, vent) is the same as in Lab D, so the Lab D 160 mm data can still be used for the design.
- **The station no longer needs the Lab D correction.** Its input showed no DC offset today, so no factor of 2 was applied. Its levels still match the Lab D data, where we *did* apply that factor, which confirms the correction was right.
- **What we cannot claim yet**: anything off axis (R4 missing), the amplifier load (impedance not measured), and whether the bare drivers are exactly where they were in Lab E (the Lab E re-check was not saved).

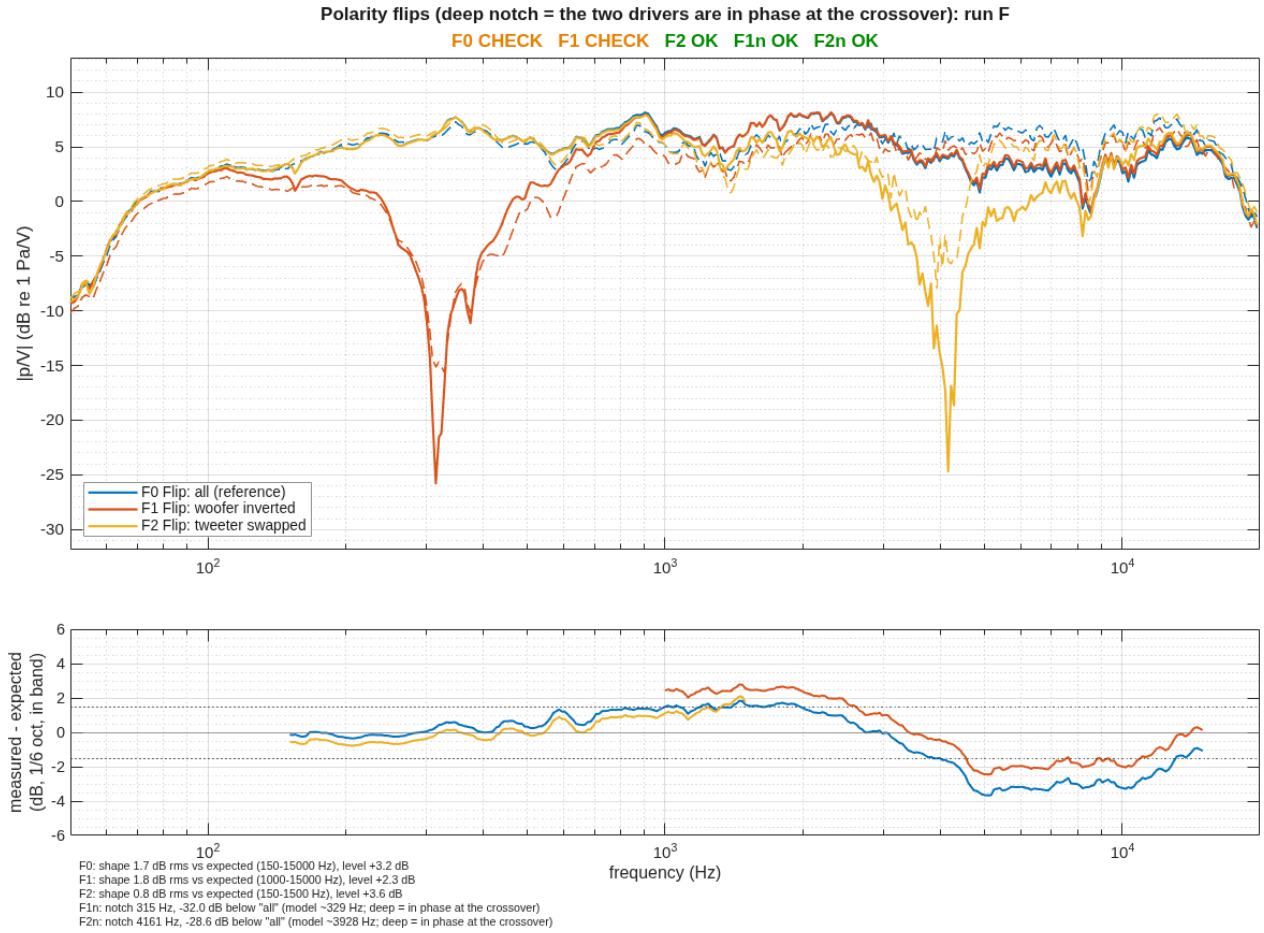


Figure 3: Polarity flips (run R3). Inverting one driver turns the sum at the crossover into a difference; the deeper the notch, the closer the two drivers were to in phase.

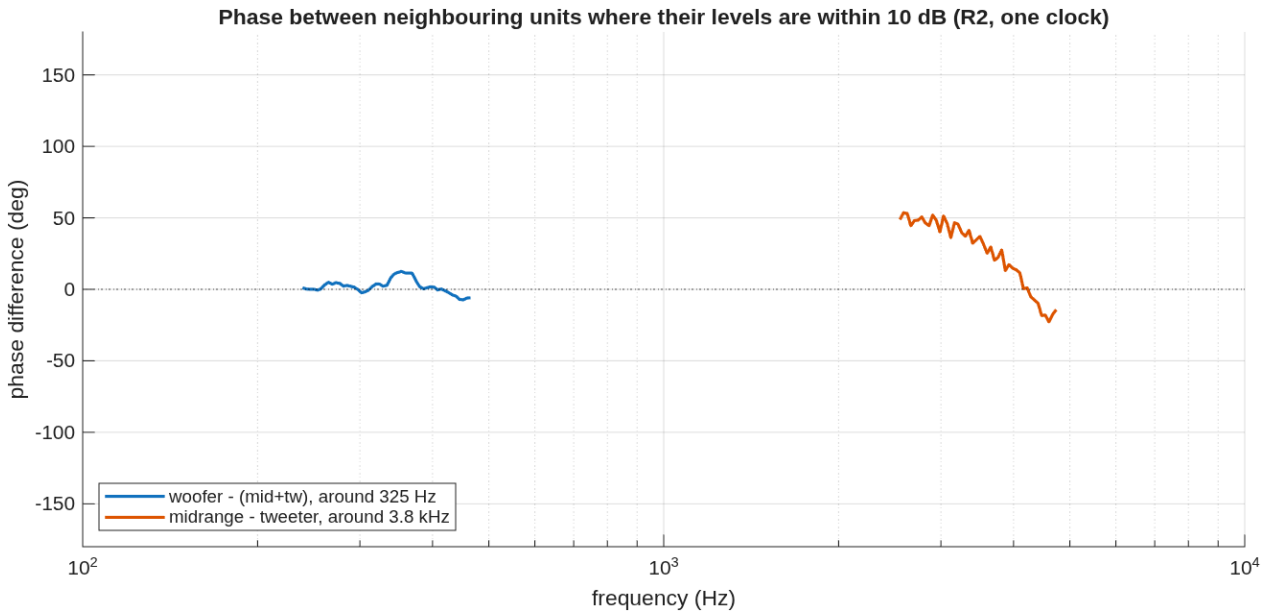


Figure 4: Phase difference between neighbouring units, taken from one run (one clock), shown only where their levels are within 10 dB of each other.

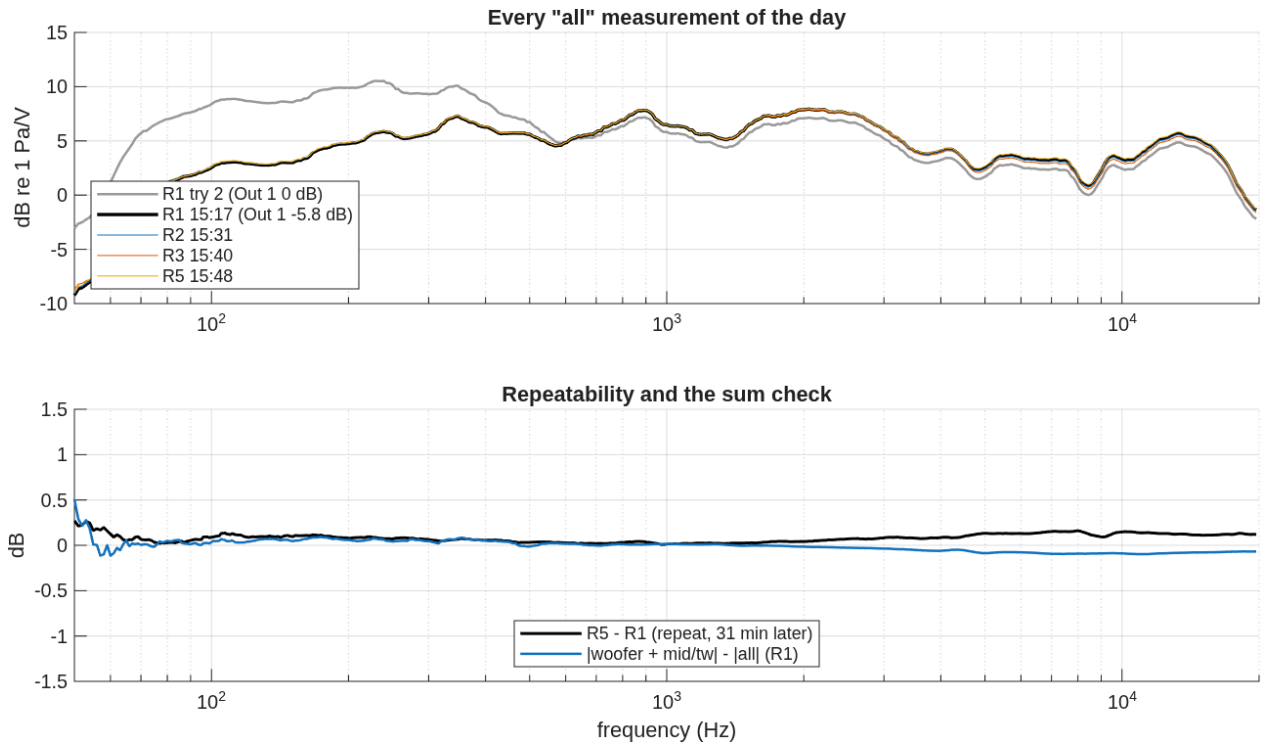


Figure 5: Top: every "all" measurement of the day. Bottom: the repeat R5 against R1, and the sum check.

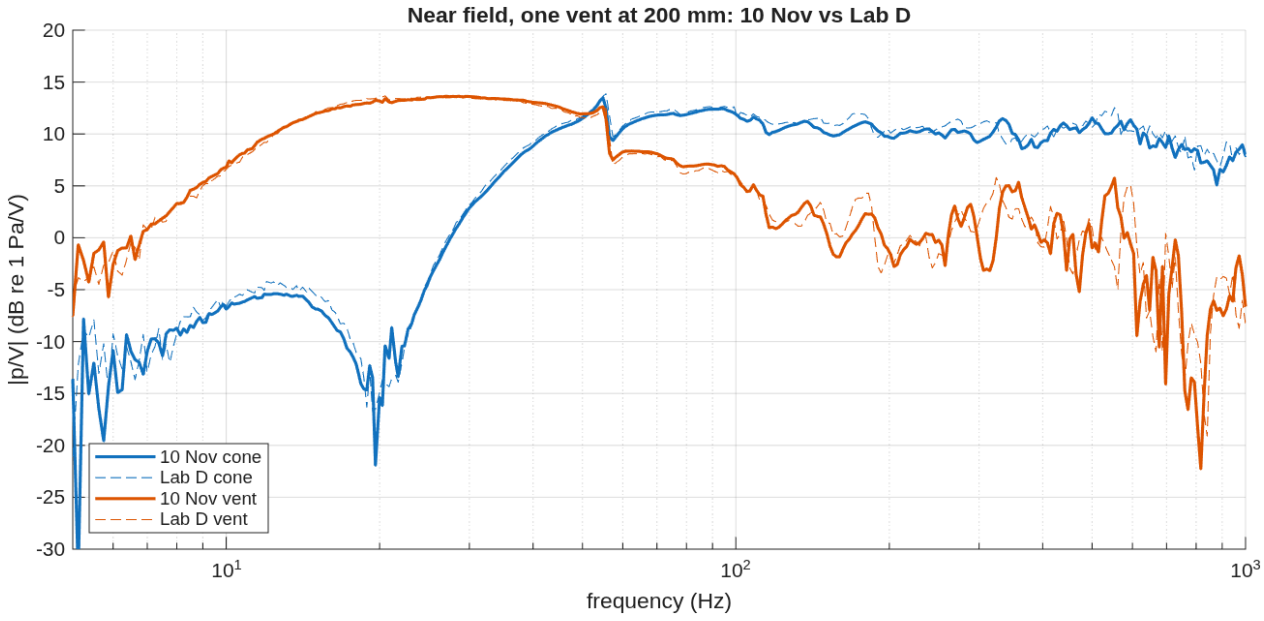


Figure 6: Near field with one vent at 200 mm, against the Lab D measurement at the same length.

4 What to improve

The design: one resistor

The tweeter's level is set by the L-pad: Rs in series and Rp across the tweeter. Run R2 measured mid, tweeter and "all" in one go, so we can take the measured units, change the pad in the board model only, and add them up again. The result is a prediction made from the real system, not from Lab E. With Appendix-1 values:

L-pad Rs / Rp (Ω)	tweeter level	spread 150 Hz–15 kHz	rms deviation
4.7 / 3.3 (as built)	0	7.2 dB	1.64 dB
3.3 / 4.7	+2.7 dB	5.6 dB	1.34 dB
4.7 / 6.8	+2.4 dB	5.3 dB	1.21 dB
4.7 / 8.2	**+2.9 dB**	**5.3 dB**	**1.20 dB**
3.3 / 6.8	+3.8 dB	6.5 dB	1.35 dB

Swap Rp from 3.3 to 8.2 Ω and keep everything else. The amplifier still sees at least 4.1 Ω. The rest of the spread is the 1–3 kHz region sitting a bit high and the 8.3 kHz dip. The dip is in the tweeter's own response, so the crossover cannot remove it. Measure the new board on Thursday (run S, then S4 + S5) and check that the tweeter step closes.

The procedure

- **Start with the placement check, and save it.** Thursday's kit has it as step P (woofer, mid, tweeter with the MiniDSP bypassed, each plotted against Lab E).
- **A graph after every measurement, with a clear OK/CHECK/BAD.** The new kit (`lab-day/kit`, `lab P`, `lab S`, …) does exactly that, plus a dashboard that always shows the whole day.
- **Write the settings down before measuring.** They go in `lab_setup.m`: Out 1 gain, amplifier, vent, series R, temperature.
- **Save the MiniDSP configuration as a preset**, so the listening test and the demo start from the same state.
- **Measure the missing pieces**: board impedance Z1–Z3 (the amplifier load, and the best check of the board) and 30° (run A).
- **Vent at 160 mm** for listening and the demo. That is the design (f_B 24.6 Hz). The 200 mm setting came from a memory of Lab D, where 200 was simply measured first.
- **The IEC room has its own amplifier**: check the woofer/mid balance there before listening (one run S). Start from the design's −4.5 dB on Out 1 there: the −5.8 dB only made up for the 1.3 dB channel difference of the chamber's amplifier.

5 What makes a loudspeaker sound good

Decades of blind listening tests (Toole, Olive and others at NRC and Harman) agree on a short list. In order of importance:

1. **A flat, smooth on-axis response.** This is the timbre: voices and instruments sound like themselves. Narrow peaks and dips matter less than broad tilts and steps. A 3 dB shelf over two octaves, like our tweeter step, is more audible than a sharp 6 dB notch. That is why the Rp change is worth more than chasing the 8.3 kHz dip.
2. **A smooth off-axis response that looks like the on-axis one, only lower.** In a room most of what reaches you after the first few milliseconds is off-axis sound reflected from the walls. If the directivity changes abruptly, for example where a large midrange that already beams hands over to a small tweeter that radiates widely, the reflections get a different colour from the direct sound. That is our open question at the 3.8 kHz crossover: the 5¼" midrange has $ka \approx 3.5$ there (beaming), while the 25 mm tweeter has $ka \approx 0.9$ (wide). The 30° run will show how big the step is. A slightly lower mid/tweeter crossover reduces it, but the tweeter resonance (1.45 kHz) limits how far down it can go.

3. **Well-integrated crossovers.** The units must add up in phase where they overlap, or you get a hole or a bump that also moves with listening height. Ours is good (Figures 3, 4).

4. **Clean, extended bass without boom.** Low-frequency extension (our f_3 is about 33 Hz) is appreciated, but a peak in the bass (too high a Q, or a room mode) is heard as boom and muddiness. The vent tuning and the room together decide this, so it is judged in the IEC room, not in the anechoic chamber.

5. **Low distortion and enough level.** This is mostly a property of the drivers and the box. For us it means not driving the woofer past its excursion limit (X_{max} 6 mm) and keeping the midrange's high-pass steep enough.

6. **Balance against the room.** A speaker that is flat on axis and whose directivity rises smoothly with frequency gives a steady downward tilt in the room, roughly −1 dB/octave. That in-room slope is what most listeners prefer. So "flat in the chamber" is the right first target: the room adds the tilt.

Flat first, then tune: how, within the course rules

The MiniDSP's PEQ, Comp and FIR blocks must stay bypassed (they cannot be exported to LTspice), so the tools are:

Knob	What it changes	Typical use
Out 1 gain (woofer)	everything below about 325 Hz against the rest	bass "fuller" +1–2 dB or "leaner"
Out 2 gain (mid+tweeter)	the same balance from the other side	keep at 0 and use Out 1
L-pad Rs/Rp	tweeter level above about 4 kHz	"bright" vs "dull": steps of about 1–3 dB
MiniDSP crossover frequency	the woofer/mid handover	300–400 Hz; the woofer starts beaming above about 500 Hz
Passive mid/tweeter values	where the mid hands over, and the shape of the overlap	lower crossover = smoother directivity
Vent length / filling	bass extension against boom	160 mm design; longer = lower f_B, less output near 30 Hz

A workable procedure: (1) measure until the system is flat (Rp = 8.2, check run S); (2) listen in the IEC room with 4–5 familiar tracks and a short questionnaire (balance, brightness, bass, clarity, imaging; 1–5 scale); (3) change *one* knob by 1–2 dB, listen again blind if possible (one person switches); (4) measure whatever you keep, so the report shows what "sounds good" means as a curve.

6 Thursday

Easiest with buttons: in MATLAB `cd lab-day/kit`, then `labgui` (every step below is a button, with the instructions, the graph and the verdict on one screen; `labgui('demo')` rehearses it without hardware). The same steps as commands:

1. Set `lab_setup.m` or the GUI's settings (Out 1 −5.8 dB, amplifier, vent 160 mm, series R).

2. `lab P`: placement against Lab E. All three should be OK before going on.

3. Put **Rp** = 8.2 Ω on the board. `lab S`, `lab S4`, `lab S5`: check the tweeter step is gone and the balance is still within ±1 dB.

4. `lab F`, `lab A` (30°), and at the Zi station `lab Z1`–`Z4`, `lab N1`, `N2`.

5. The dashboard shows everything as you go. Copy `data/<date>/` (including its `figures/`) to the stick (GUI: *Copy day to USB*).